

Vishvesh Singh Pal

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Profile Summary

Cloud Native Engineer and DevOps practitioner specializing in architecting highly scalable, geo-distributed infrastructure and automating CI/CD pipelines. Proficient in Infrastructure as Code (IaC), zero-trust networking, and Kubernetes orchestration. Experienced in building and maintaining robust observability stacks and deploying AI/ML workloads to production environments.

Skills

Cloud & Infrastructure: AWS (ECS, ECR, EC2), GCP (Cloud Run, GCS, Vertex AI), Oracle Cloud, Terraform, Ansible.

Containerization & Orchestration: Kubernetes (K3s), Docker, Helm, ArgoCD.

Networking & Security: WireGuard, Tailscale, Headscale, Flannel CNI, Cloudflare Tunnels, Traefik, Nginx, Authentik.

Observability & Data: Prometheus, Grafana, InfluxDB, Kafka, RabbitMQ/SQS.

Languages & Frameworks: Python, Bash, SQL, FastAPI, LangChain.

Professional Experience

YASH Technologies, Indore –Trainee

Jan’25 – Present

Key Projects:

- Automated cloud infrastructure provisioning on Google Cloud using Terraform, configuring highly-available GCS (Google Cloud Storage) for secure storage of large-scale audio datasets and pipeline artifacts.
- Developed a high-throughput, low-latency serverless audio transcription and analysis pipeline; containerized and deployed to Google Cloud Run using Vertex AI and LangChain.
- Engineered a robust, automated CI/CD pipeline with GitHub Actions for container image building, testing, and pushing to Google Cloud, ensuring reliable, zero-downtime application updates.

CodeWalnut , Bangalore – Fullstack Intern

June’24 – Jan,25

Key Projects:

- Architected and scaled cloud environments on AWS ECS, leveraging Terraform for Infrastructure as Code (IaC) management, achieving 100% consistency and configuration parity across Dev, Staging, and Production environments.
- Reduced infrastructure risk by implementing a GitOps workflow for version-controlled, automated deployments, effectively eliminating manual intervention and reducing configuration drift.
- Engineered a robust CI/CD ecosystem using GitHub Actions and Amazon ECR to automate the build-to-deploy lifecycle, significantly accelerating release velocity.
- Implemented a comprehensive observability stack using Prometheus, Grafana, and CloudWatch Logs to monitor system health, which resulted in a measurable reduction in Mean Time To Detect (MTTD) via real-time dashboards and proactive alerting.

Projects

Geo-Distributed Self-Hosted K3s Cluster | [\[GitHub Link\]](#)

- Orchestrated a hybrid, geo-distributed Kubernetes (K3s) cluster across Oracle Cloud instances and on-premises Proxmox nodes.
- Deployed a self-hosted WireGuard mesh network via Headscale to bridge multi-site infrastructure, configuring Flannel CNI over Tailscale for encrypted pod-to-pod communication.
- Deployed a GitOps pipeline via ArgoCD for autonomous workload management, ensuring scalable resource allocation for LLM inference pipelines and media services.

AI Child Safety DNS Firewall

- Engineered an AI-driven Smart DNS Firewall for real-time content analysis on unknown domains, leveraging Amazon Bedrock (Nova models) for classification.
- Developed an automated ingestion pipeline using Amazon SQS to asynchronously queue DNS requests, ensuring zero impact on user latency during real-time evaluation.
- Engineered a closed-loop security system where BIND configurations dynamically update based on AI safety scores.

Real-Time Transport Monitoring (Delhi OTD) | [\[GitHub Link\]](#)

- Deployed a scalable, event-driven architecture on Kubernetes, utilizing MQTT and Kafka for continuous ingestion of public transit telemetry.
- Configured InfluxDB for time-series data storage and constructed a CI/CD pipeline using GitHub Actions and Helm charts for seamless deployment management.

Education

B.S. in Data Science and Applications
Indian Institute of Technology, Madras

2021-2025